

CTRL–C2000 control card

Rev E0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | TMS320F28388D common controller for PV–P75/100/110 and DAB–D60

01 Power regulators

+24V -> LMR38020 5 V -> TPS62133 3.3 V; PG chain enables TPS62130 1.2 V and TPS7A2033 3.3 V analog (VDDIO first, then VDD and VDDA)

File: sheets/01_power.kicad_sch

02 Supervisors, reset, clock, boot

TPS3703 windows on 3.3 V and 3.3 V analog, UV supervisor on 1.2 V -> XRSn; 25 MHz CMOS clock into X1; boot straps GPIO72/GPIO84 (flash by default)

File: sheets/02_reset.kicad_sch

03 TMS320F28388D (ZWT 337–ball)

All 337 balls, pin table extracted from SPRSP14E (gen/data/f28388d_pins.csv); net per ball from the pin plan (gen/data/ctrl_c2000_pin_plan.csv)

File: sheets/03_mcu.kicad_sch

04 MCU supply decoupling

SPRSP14E 7.10.1.5: 0.1 uF per VDDIO/VDD3VFL/VDDOSC ball, 2.2 uF per VDDA ball, -22 uF total on VDD + one 56 R load, 22 uF per VREFHI (16–bit mode)

File: sheets/04_decoupling.kicad_sch

05 ADC reference, VMID, slow mux

REF5025E 2.5 V for VREFHIA–D; OPA2388 buffers VMID 1.25 V (diff–amp offset) and VBIAS 2.5 V (ID/NTC bias); TMUX1208 for 4 NTC + 4 ID; 24 V monitor; AGND star

File: sheets/05_adcref.kicad_sch

06 Fault receivers, enable, EN drivers

Cn_EN = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only while CHB_OK, MCU enable and RDY; EN drivers with readback; fail–safe FLT receivers

File: sheets/06_trip.kicad_sch

07 Cell port C1

CELL contract 2x20: 8 PWM RS–422 drivers, 3 dual–range analog inputs, RDY, ID, NTC; gate–driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/07_cell1.kicad_sch

08 Cell port C2

CELL contract 2x20: 8 PWM RS–422 drivers, 3 dual–range analog inputs, RDY, ID, NTC; gate–driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/08_cell2.kicad_sch

09 Cell port C3

CELL contract 2x20: 8 PWM RS–422 drivers, 3 dual–range analog inputs, RDY, ID, NTC; gate–driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/09_cell3.kicad_sch

10 Cell port C4

CELL contract 2x20: 8 PWM RS–422 drivers, 3 dual–range analog inputs, RDY, ID, NTC; gate–driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/10_cell4.kicad_sch

11 Port–board analog inputs

PORT contract 2x13: VA, VAX, VB, VBX, IA, IB, AUX1, AUX2 isolated–amplifier pairs, biased open–safe, through difference amp + RC + clamp; ID: +24V out through an eFuse

File: sheets/11_port.kicad_sch

12 SYS–IO–AUX connector

80–way SYS contract: +24 V logic, +24V_GD (channel B), CAN, SCI, SPI, I2C, DI/DO, fault latch, GATE_EN, CHB_OK, watchdog, reset, fans, Ethernet MDI + centre tap

File: sheets/12_sys.kicad_sch

13 Ethernet PHY DP83822I

MII to the CM EMAC (GPIO105–124, mux 14), own 25 MHz oscillator, 49.9 R MDI terminations to AVD; MDI pairs, LEDs and the centre–tap supply ETH_CT go to SYS

File: sheets/13_eth.kicad_sch

14 JTAG, UART, EEPROM, LEDs, SDFM

cTI–20 JTAG, CM debug UART, 24LC256 calibration EEPROM (0x57), status LEDs, ERRORSTS, 8–channel SDFM expansion header (ECO–02 / ECO–09)

File: sheets/14_debug.kicad_sch

Protection without firmware: FLT_LATCH_N -> INPUTXBAR1 -> TZ1; each cell’s FAULT (INPUTXBAR2–5) and AN1 CMPSS window -> TRIP4/5/7/8; port OV (ADC PPB on VA/VB, CMPSS backup), open inputs (CMPSS low), IA/IB SC (CMPSS) and OC (PPB), P_ID (PPB) -> TRIP9 for every cell; E–stop -> INPUTXBAR6 -> TRIP6. CMPSS reference = VDAC = REF5025E; firmware nulls each channel at zero input at start–up; trip bands in the pin plan, asserted against sim/out/pv_control/control_spec.json. EN pair = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only with CHB_OK, MCU enable and RDY; EN readback on GPIO100–103. +24V_GD feeds only the cell eFuses.

Rails: 24 V -> 5 V (LMR38020) -> 3.3 V (TPS62133); its PG enables 1.2 V (TPS62130) and 3.3 V analog (TPS7A2033), so VDDIO leads VDD and VDDA (SPRSP14E table 7–1). Three TPS3703 hold XRSn (windows on 3.3 V and 3.3 V analog, UV on 1.2 V), every tolerance corner asserted. REF5025E runs from the analog rail, so VREFHI and VDAC never exceed VDDA.

Pin plan with mux positions and trip bands: gen/data/ctrl_c2000_pin_plan.csv (checked against SPRSP14E table 6–1 on every build). Pin table: gen/data/f28388d_pins.csv (from the PDF).

SYS side: inputs via 74LVC244A / 74LVC1G17 (loff); I2C pull–ups and EEPROM 0x57 here; ETH_CT = PHY AVD; PORT +24V via its own eFuse.

Not bench–validated. Values marked calculated are not simulated.
Sheet 1
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: / File: CTRL–C2000.kicad_sch		
Title: CTRL–C2000 control card – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 1/15

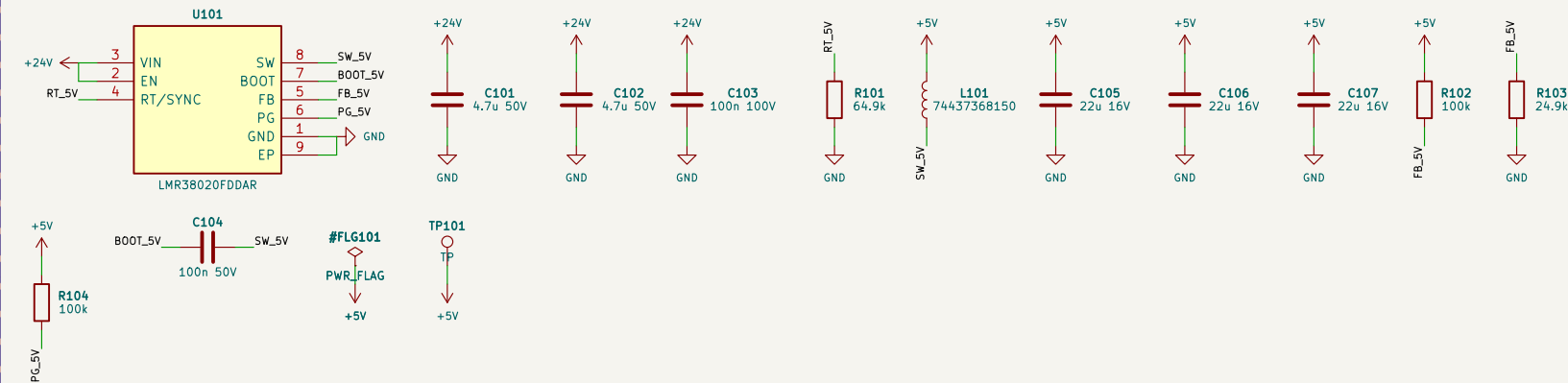
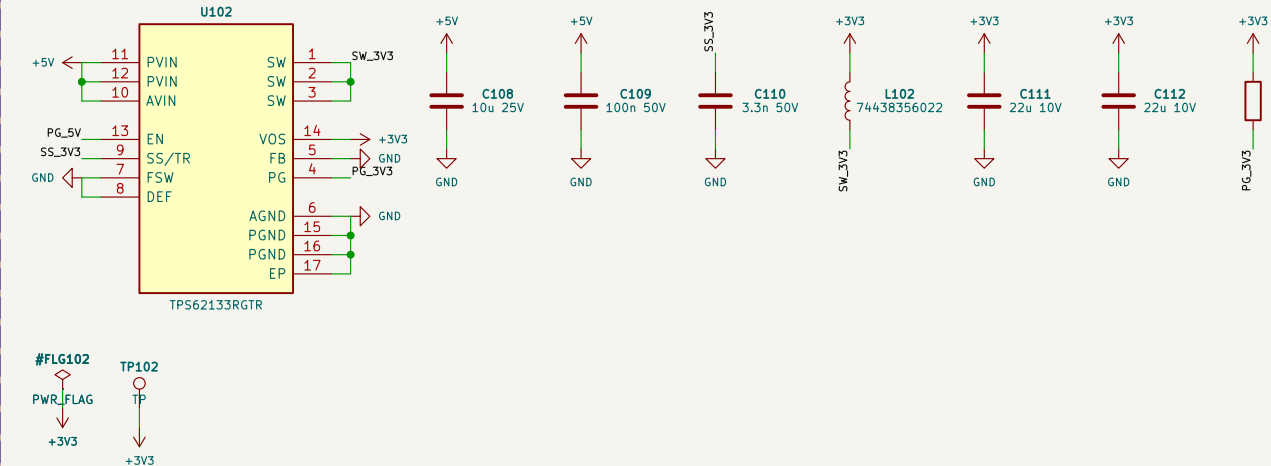
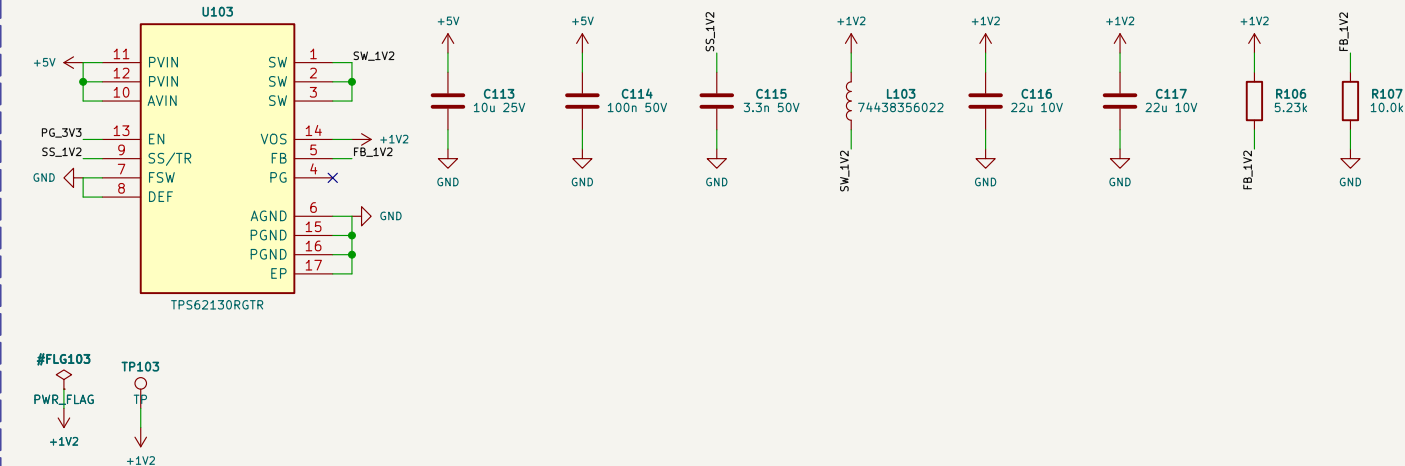


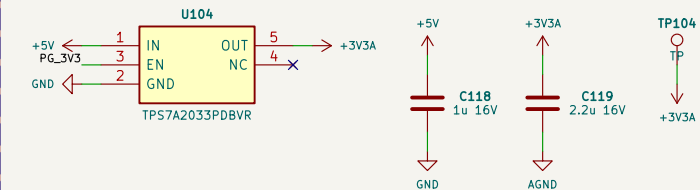
Table 9—load run 400 kHz; V = 2 A; I_V = 15 V, 3x2x2 UFLB, RTF100 / RFBF 29.4kΩ → 5.02 V, RT 64.9kΩ.
Load (calculated): 3.3 V 1.5 A + 1.2 V 0.48 A + 3.3 V analog 0.1 A → about 1.2 A at 5 V.
EN tied to VIN (datasetset loads). PGV5 enables the 3.3 V converter.
Output capacitance check (SNVS/C40C, calculated): total on +5V = 3x22 + 2x10 + 1 + 0.3 = 87 μF nominal,
96 μF at +10 %; 1.45x the 66 μF design value, inline the 9.2-2.5 limit (10x; design or 1000 μF) → no extra
capacitors needed. Inductor ripple current ΔI_L=ΔI_C/Q=0.15 A/0.5 ms = 0.42 A charge current;
downstream loads held off by the PG chain, incl. ILIMIT 18 A min; OCP blanking 18 ms → no hiccup;
inductor: LMIN = 0.25 x 5 V / 400 kHz = 3.1 μH (eq. 11); < 15 μH; ripple 0.64–0.68 A at 21.6–26.4 V in
(> 10 % of 2 A); peak 1.54 A (1.62 A at L – 20 %) (< IH-LIMIT 2.6 A min | ISAT 5.55 A, Bode not measured).



Fixed 3.3 V; FB to AGND per datasheet. FSW low (2.5 MHz), 2.2 uH, 2x22 uF (table 9-1). SS 3.3 nF (typical application). PG_3V3 releases the 1.2 V and 3.3 V analog rails. Worst case 3.221–3.396 V (power-save mode, SLVSAG7F p5) inside VDDIO 3.14–3.47 V.



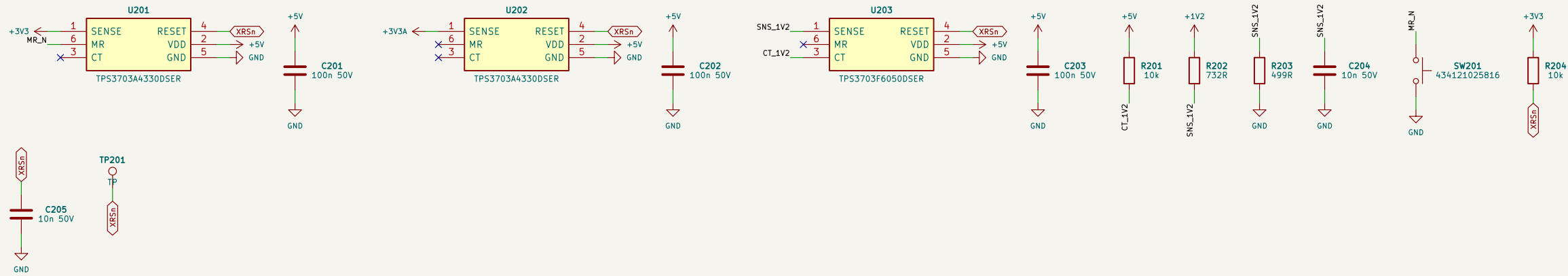
VOUT = 0.8 V x (1 + 5.23k/10.0k) = 1.218 V (eq. 6), 0.1 % 10 ppm/K: worst case 1.1872–1.2558 V (power-save VFB 781.6–822.4 mV, FB leakage, 0.1 % (line/load) inside VDD 1.14–1.26 V (CSR=01). Enabled by PG_3V3: VDD rises after VDDIO (SPRSP14E table 7-1). VDD bulk on the decoupling sheet.



VDDA, op amps, reference, mux. Enabled by PG_3V3 so VDDA never leads VDDIO.
Load (calculated): 26 op-amp channels x 2.6 mA max + IDDA 15 mA + REF and its 4.2 mA load + VBIAS
+ 12 open-input bias resistors to +3V3A (16 mA) = about 110 mA of 300 mA.

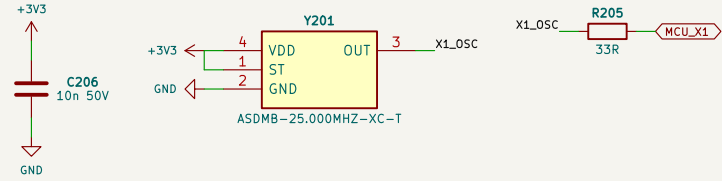
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 2/15

SUPPLY SUPERVISORS (OPEN-DRAIN RESETS WIRED TO XRSN), CSR-01



SPRSP14E 7.10.1.2.4: VDDIO and VDD need an external supervisor (the internal POR trips below range). Worst case at every corner (asserted on each build, MCU window 3.14–3.47 V / 1.14–1.26 V):
+3V3 A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2209$; OV 3.408–3.456 $> \text{regulator max } 3.3958$ V
+3V3A A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2505$; OV 3.408–3.456 $> \text{regulator max } 3.3495$ V
+1V2 F6050: trips ≥ 1.1454 , releases $\leq 1.1829 < \text{regulator min } 1.1872$; regulator max 1.2558 V
1.2 V is UV-only: a $\pm 4\%$ window cannot hold the TPS62130 power-save spread; its maximum is the regulator arithmetic above. Delay: CT open 10 ms (A parts), CT to VDD via 10k 20 ms (F part). VDD = +5V so the outputs stay defined while 3.3 V collapses. XRSn pull-up 10k, 10 nF; SYS_RST_N and the JTAG RESET are open drain on the same node. Reset button on MR of the 3.3 V part (100k internal pull-up).

25 MHZ CLOCK (X1)



SPRSP14E 7.10.3.2.1: X1 from an oscillator 10–25 MHz, tr/ff ≤ 6 ns, 45–55 %; X2 open (figure 7–11). No crystal, so VSSOSC goes to board ground (table 6–12). 33 R source termination.

BOOT MODE (GPIO72 = PIN 1, GPIO84 = PIN 0)

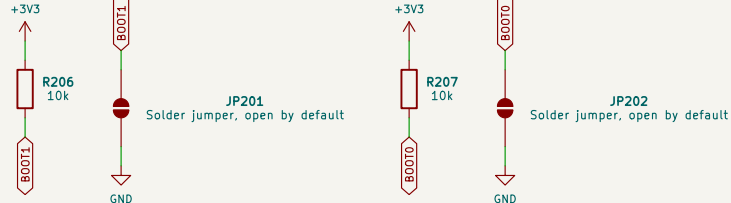


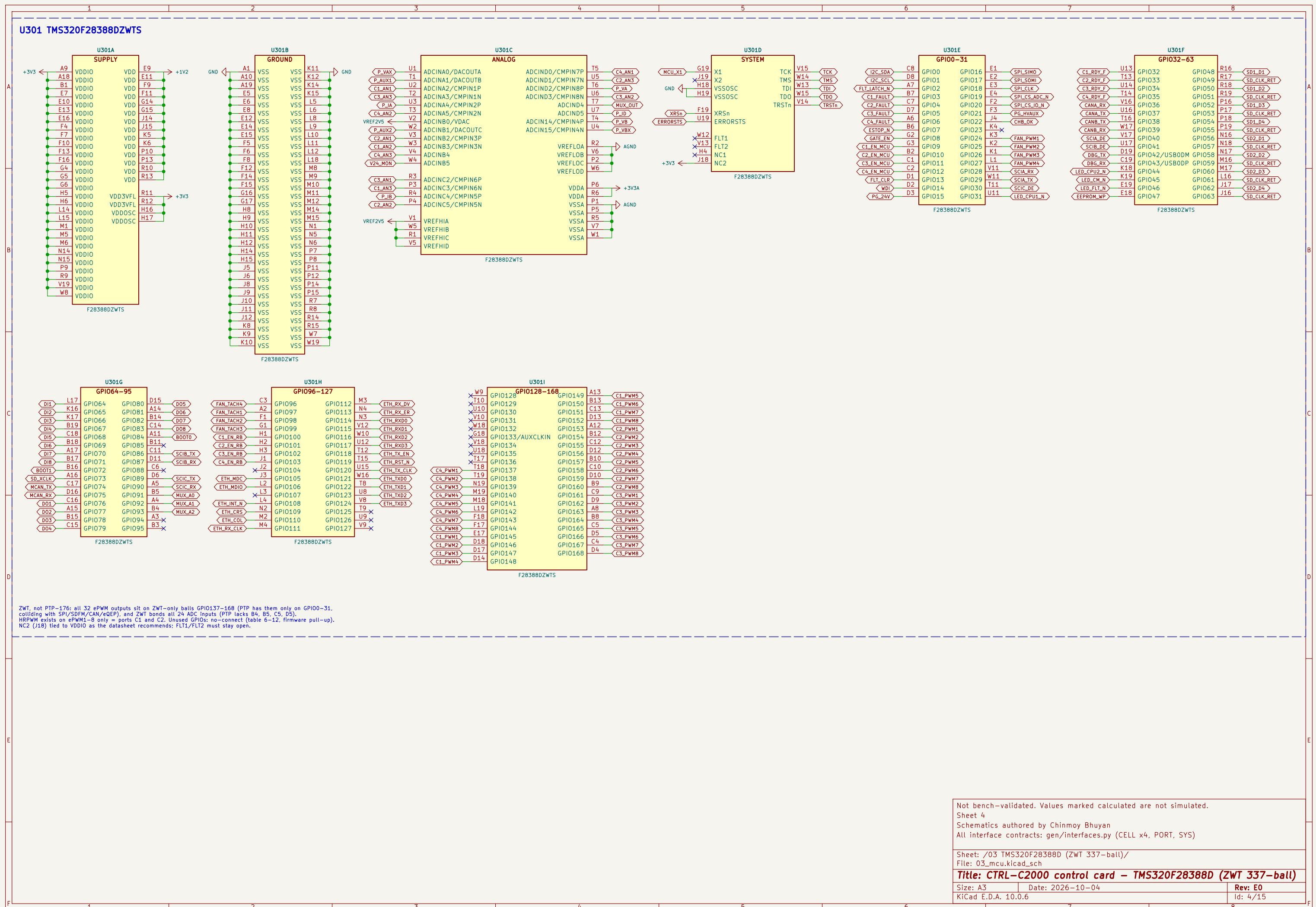
Table 8–18: 11 = flash (straps open), 01 = SCI, 10 = CAN, 00 = parallel I/O. Close JP to pull a pin low. th(boot-mode) 1.5 ms; nothing else drives these balls.

Not bench-validated. Values marked calculated are not simulated.
Sheet 3
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

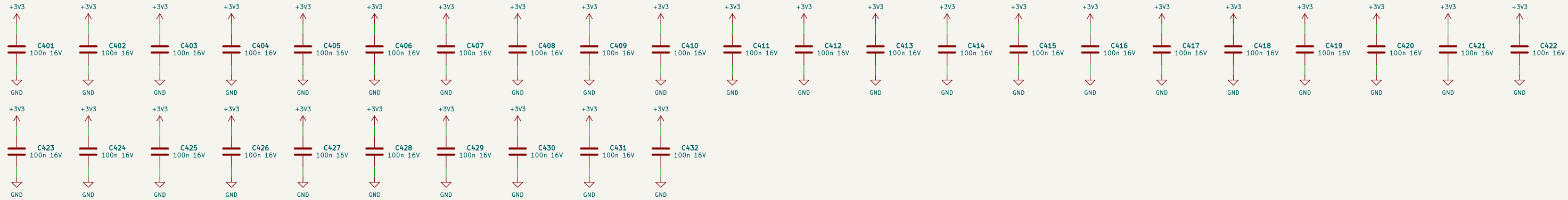
Sheet: /02 Supervisors, reset, clock, boot/
File: 02_reset.kicad_sch

Title: CTRL-C2000 control card – Supervisors, reset, clock, boot

Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 3/15

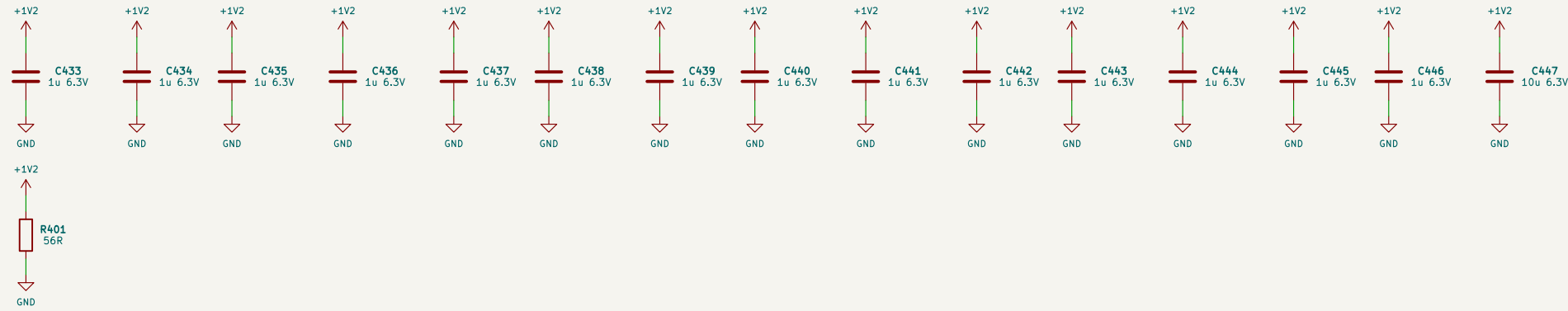


VDDIO (28 BALLS) + VDD3VFL (2) + VDDOSC (2): 0.1 UF EACH



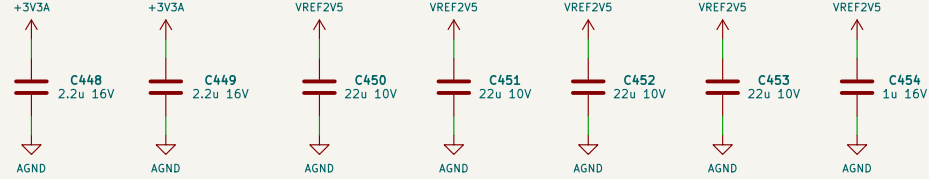
Configuration 1 of 7:10.1.3.2.

VDD 1.2 V (14 BALLS)



14 x 1 uF + 10 uF = 24 uF, 10 % parts: >= 21.6 uF (CVDD TOTAL 20 min / 22 typ. max tolerance 20 %).
56 R load consumes the VDD3VFL-to-VDD leakage current (table 6-4 note).

VDDA, VREFHI AND VDAC



VDDA 2.2 uF per ball to VSSA. VREFHIA-D: 22 uF each to VREFLO (16-bit mode).
VDAC (ADCINB0, V2) = CMPSS DAC reference: >= 1 uF at the ball (table 6-1).
All on VREF2V5: 4 x 22 + 1 + 1 uF = 90 uF, 99 uF at +10 %: inside the REF5025E 1-100 uF range (summed from the design on every build).

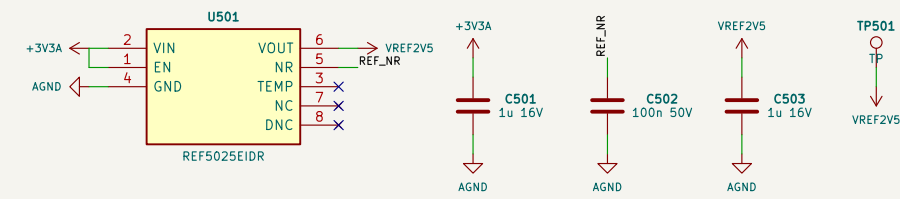
Not bench-validated. Values marked calculated are not simulated.
Sheet 5
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /04 MCU supply decoupling/
File: 04_decoupling.kicad_sch

Title: CTRL-C2000 control card - MCU supply decoupling

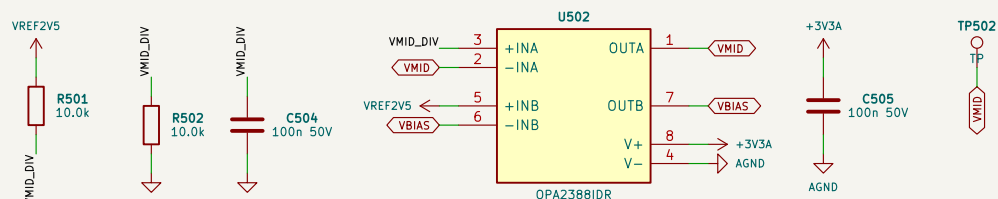
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 5/15

2.5 V ADC AND CMPSS DAC REFERENCE



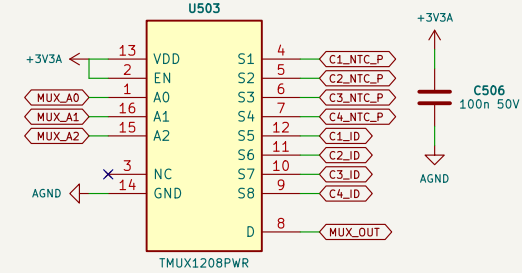
Powered from +3V3A (VIN >= VOUT + 0.2 V) so VREFHI and VDAC can never exceed VDDA (SPRSP14E 7.11.2.3.5, 7.11.3.1.4). Also the CMPSS DAC reference (VDAC, INT-05); thresholds ratiometric with the ADC.
Load 4.2 mA of +/-10 mA; VDAC 8 x 6 k (3.3 mA) + VREFHI 4 x 190 uA + VMID divider (asserted).
NR 0.1 uF class 1; 1 uF HF at VOUT; VREFHI and VDAC capacitors are on the decoupling sheet.

VMID AND VBIAS BUFFERS



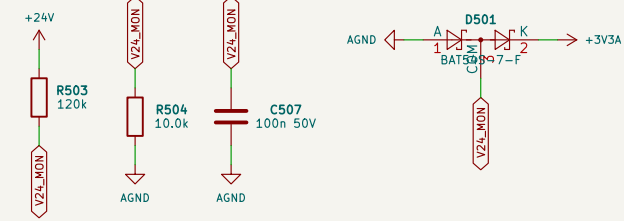
VMID = VREF/2 is ratiometric: a zero differential input reads mid-scale whatever the reference error.
VBIAS = VREF buffered, so ID and NTC readings are ratiometric and the reference itself is not loaded.

SLOW-SIGNAL MUX -> ADCIND4



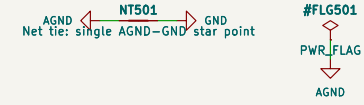
S1-S4 = C1-C4 NTC, S5-S8 = C1-C4 ID (24 ADC balls cannot take 20 fast channels + 9 slow ones).
Each input has its own 100 nF on the cell sheet; the mux only switches between charged caps.

24 V MONITOR -> ADCINB5



120k/10k: 24 V reads 1.85 V, full scale 32.5 V: 100 nF filter.
+24V is live before VDDA: the BAT54S holds the ball at VDDA + VF (7.10.1.4.2).

AGND - GND STAR POINT



Single connection of the analog return to board ground (layout: under U301).

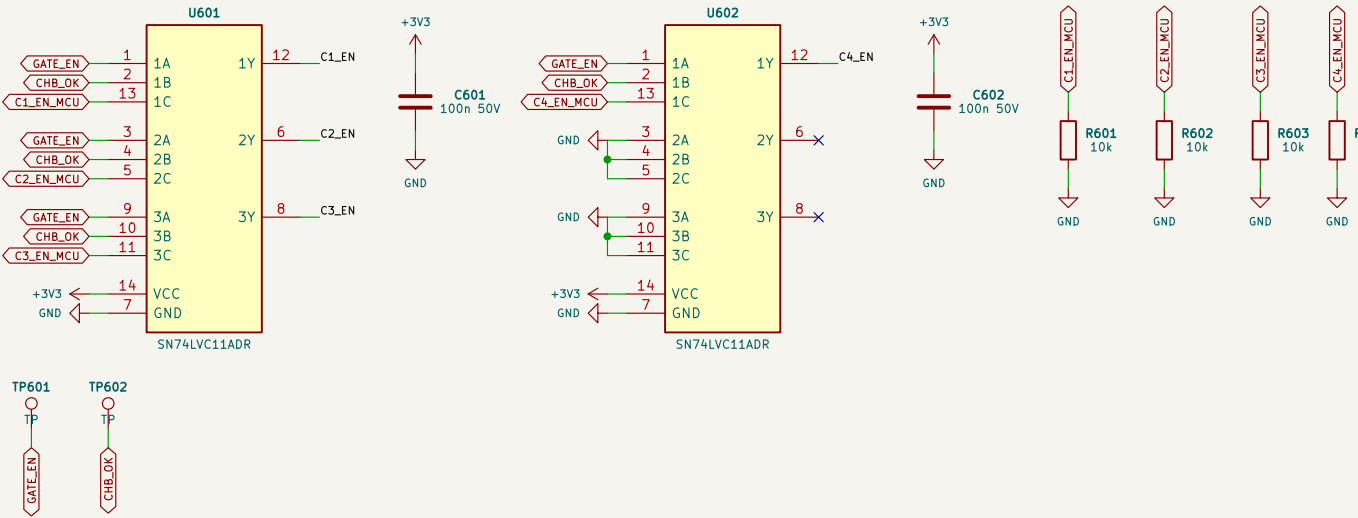
Not bench-validated. Values marked calculated are not simulated.
Sheet 6
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /05 ADC reference, VMID, slow mux/
File: 05_adcref.kicad_sch

Title: CTRL-C2000 control card - ADC reference, VMID, slow mux

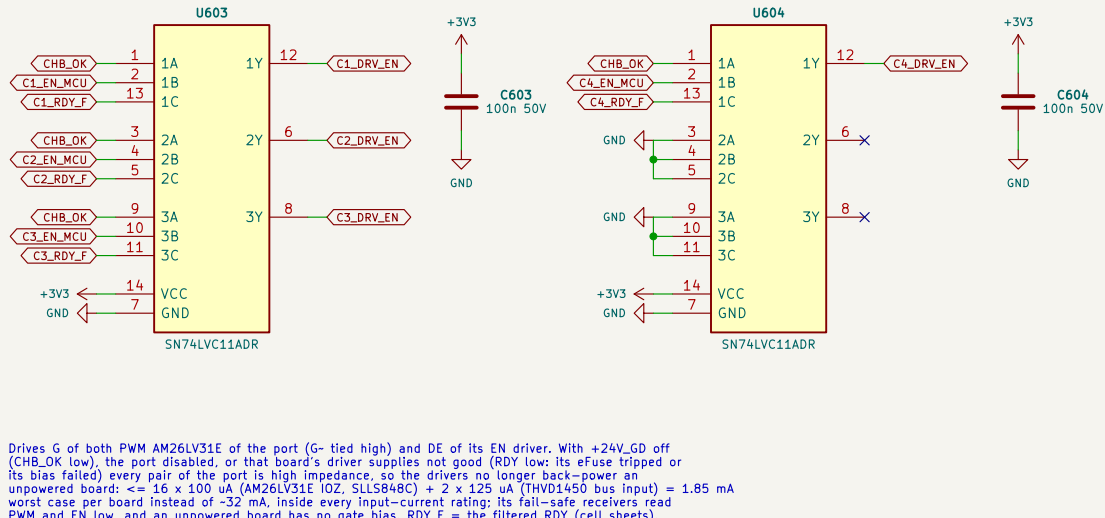
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 6/15

HARDWARE GATE-ENABLE AND (BOTH SAFETY CHANNELS)



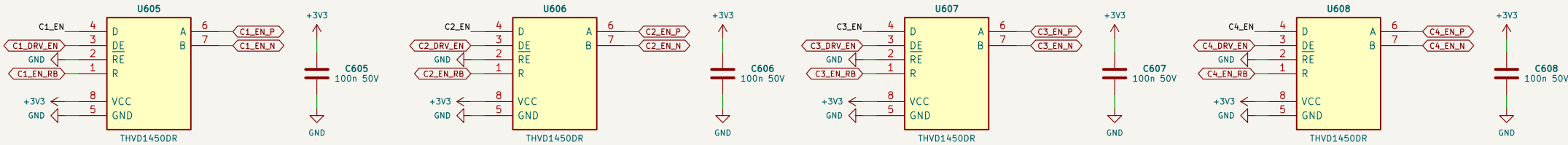
Cn_EN = GATE_EN (channel A) AND CHB_OK (channel B) AND Cn_EN_MCU, one 3-input gate per port; firmware alone can never raise an EN pair. Channel B also removes +24V_GD (gate-driver bias) on SYS. 10k pull-downs hold each MCU enable low in reset; GATE_EN and CHB_OK are pulled low on the SYS side. Spare gates: inputs to GND, outputs open.

LINE-DRIVER ENABLE CN_DRV_EN = CHB_OK AND CN_EN_MCU AND CN_RDY (INT-15)



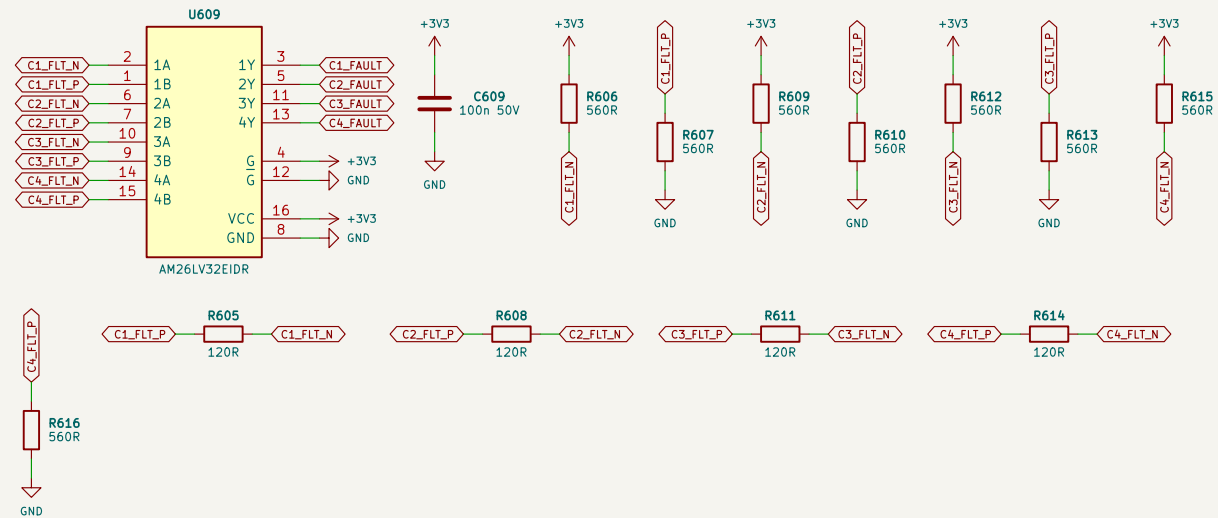
Drives G of both PWM AM26LV31E of the port (G- tied high) and DE of its EN driver. With +24V_GD off (CHB_OK low), the port disabled; or that board's driver supplies not good (RDY low: its eFuse tripped or its bias failed) every pair of the port is high impedance, so the drivers no longer back-power an unpowered board: $\leq 16 \times 100 \mu A$ (AM26LV31E IOZ, SLLS848C) + $2 \times 125 \mu A$ (THVD1450 bus input) = 1.85 mA worst case per board instead of -32 mA, inside every input-current rating: its fail-safe receivers read PWM and EN low, and an unpowered board has no gate bias. RDY_F = the filtered RDY (cell sheets).

EN LINE DRIVERS WITH READBACK (C1-C4)



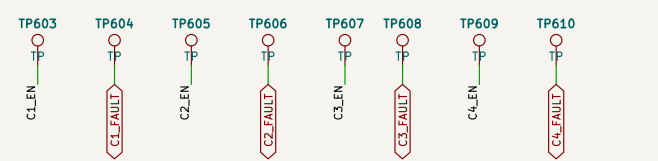
THVD1450 per port: D = Cn_EN, DE = Cn_DRV_EN, A/B = EN_P/EN_N, [VOD] ≥ 2 V into 100 R (SLLSEY3E 7.7). Receiver always on (RE- low): R = the pair's state at the CTRL end $\rightarrow$ Cn_EN_RB on a spare GPIO, so a stuck AND gate or driver is visible (INT-22: start-up and proof-test sequence in the pin plan).

FLT RECEIVERS (C1-C4), FAIL-SAFE



Pair wired swapped (A = FLT_N, B = FLT_P): Cn_FAULT is high = fault. 560R bias + 120R termination put +0.32 V across an open/unplugged pair ($\rightarrow$ VIT+ 0.2 V) $\rightarrow$ fault; the AM26LV32E rev E only has open-circuit (not terminated) fail-safe, hence the external bias. Healthy: driver -2 V $\rightarrow$ low, Cn_FAULT $\rightarrow$ INPUTXBAR2-5 $\rightarrow$ ePWM X-BAR $\rightarrow$ TRIP4/5/7/8 (no firmware in the trip path).

TEST POINTS



Gated EN (after the AND) and receiver FAULT outputs, one pair per port.

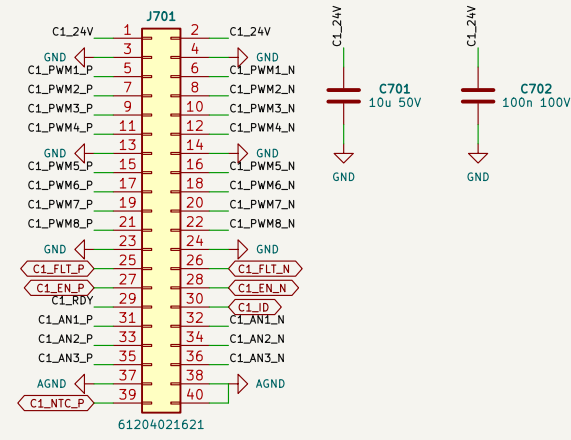
Not bench-validated. Values marked calculated are not simulated.
Sheet 7
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /06 Fault receivers, enable, EN drivers/
File: 06_trip.kicad_sch

Title: CTRL-C2000 control card - Fault receivers, enable, EN drivers

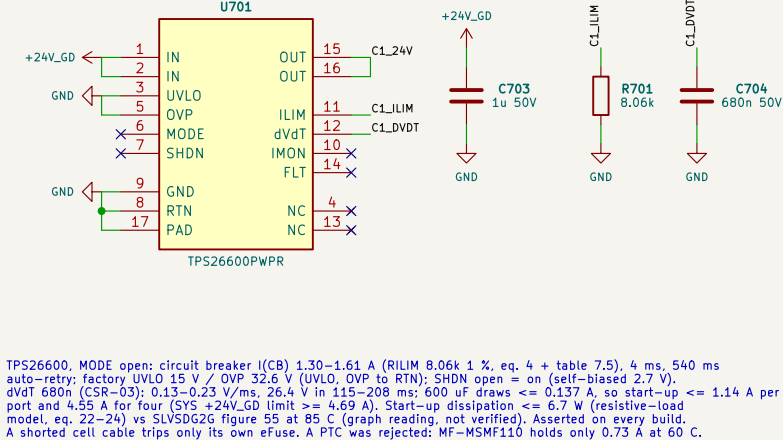
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 7/15

C1 CONNECTOR (CELL CONTRACT)



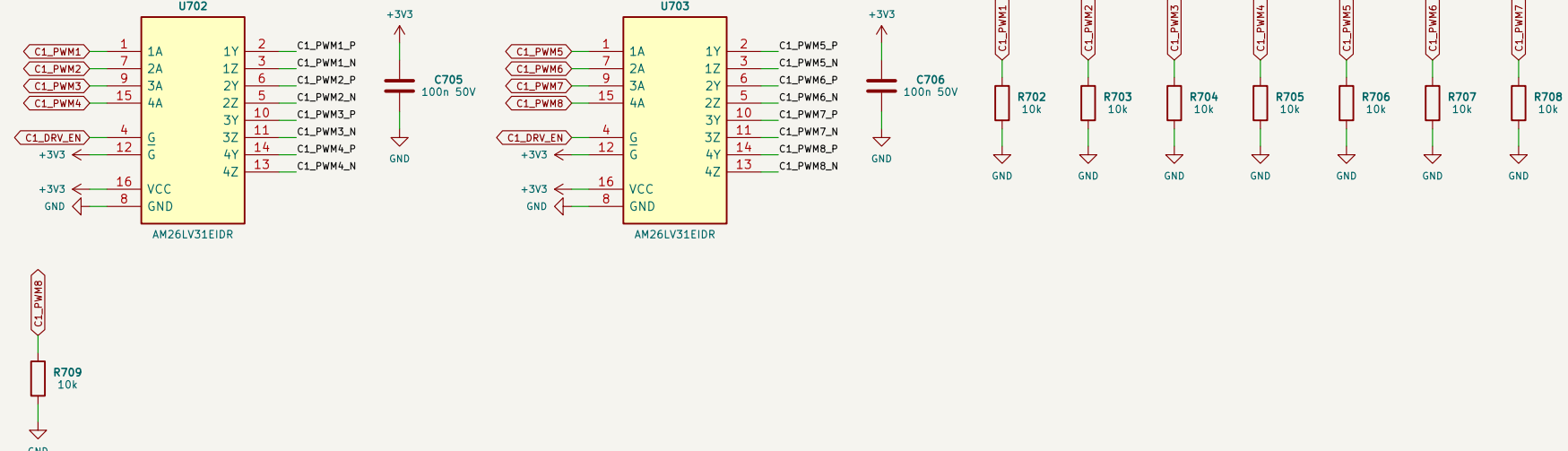
Pin map = Interfaces.CELL. Pins 1-2 carry +24V_GD only (gate-driver bias, channel B switched).
NTC_N returns to AGND here; FLT/EN gains go to the trip sheet.

+24V_GD EFUSE FOR C1 (<= 1.0 A CONTINUOUS, <= 600 uF)



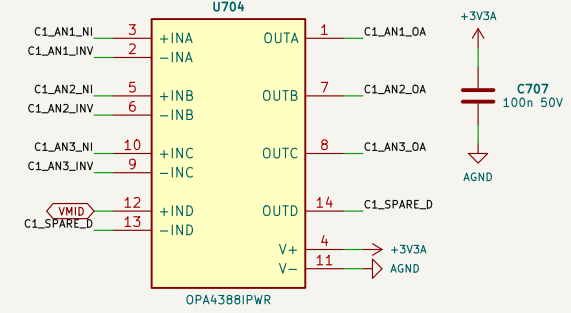
TPS26600, MODE open: circuit breaker [(CB) 1.30-1.61 A (RILIM 8.06k 1 %, eq. 4 + table 7.5), 4 ms, 540 ms auto-retry; factory UNLO 15 V / OVP 32.6 V (UVLO, OVP to RTN); SHDN open = on (self-biased 2.7 V).
dVdT 680k (CSR-D3): 0.13-0.23 V/ms, 26.4 V in 115-208 ms; 600 uF draws <= 0.137 A, so start-up <= 1.14 A per port and <= 55 A for four (SYS +24V_GD limit >= 4.69 A). Start-up dissipation <= 6.7 W (resistive-load model, eq. 22-24) vs SLVSDG2G figure 55 at 85 C (graph reading, not verified). Asserted on every build.
A shorted cell cable trips only its own eFuse. A PTC was rejected: MF-MSMF110 holds only 0.73 A at 60 C.

PWM LINE DRIVERS (EPWM1-4)



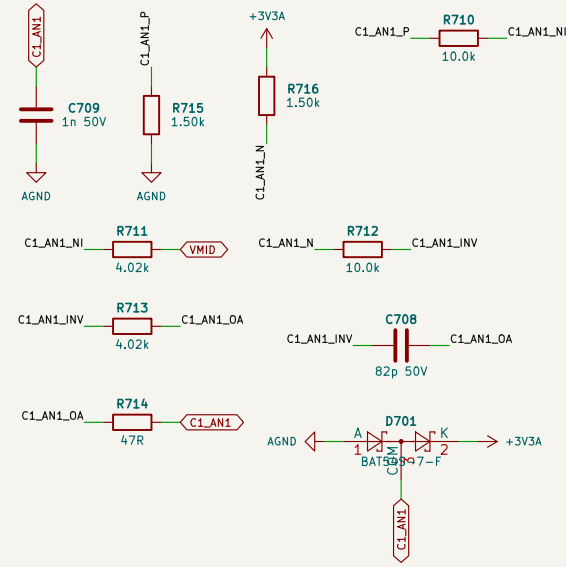
PWMx (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs.
G = CL1DRV_EN, G=high: outputs high impedance unless CHB_OK, CL1EN_MCU and RDY are high (N1-15).
ePWM1-4 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

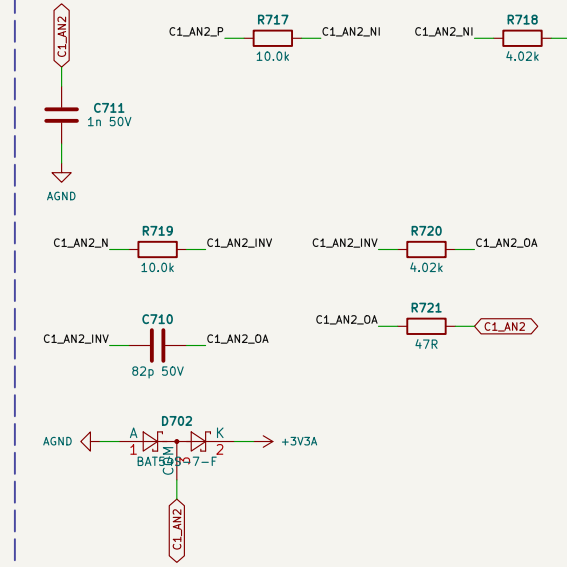


Difference amp per pair: $V_{ad} = V_{MD} 1.25 V + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$.
C1 || Rf anti-alias pole 483 kHz, 47 R + 1 nF CG; ADC charge bucket (ACQPS >= 320 ns at 12 bit).
BAT5MS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated.
Accepts on any AN pair: differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with AN_N = AGND at the source (temperatures) -> 1.25-2.46 V, Op-amp inputs 0.9-1.75 V (RMID). Source resistance adds to the 10.0k: keep R <= 100 R (1 % gain) or calibrate it out.
AN1 = inductor current (INT-12, one constant); PVCELL-25 IEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11k A -> 8.0k mV/A at the ADC. 0.0759 A/LSB; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it.
DAB-D60 (C1/C2): AN1 5.009 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMD follower.

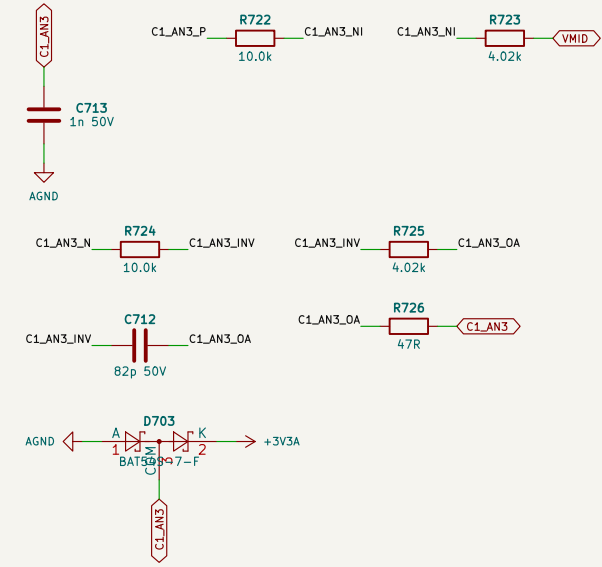
C1_AN1 -> ADCINA2 / CMPIN1P



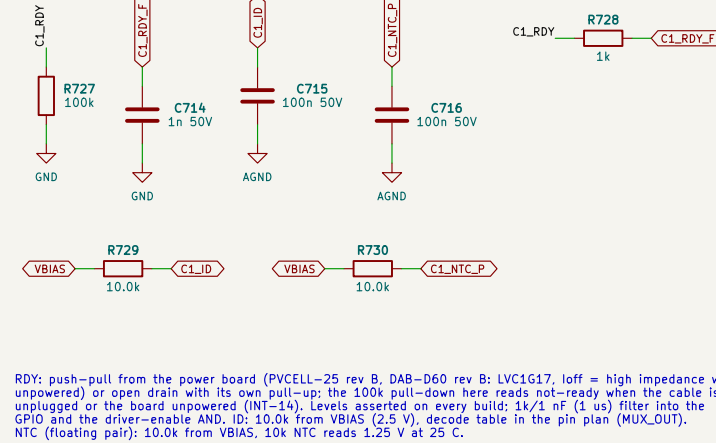
C1_AN2 -> ADCINB3



C1_AN3 -> ADCINC3



RDY, ID, NTC



RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (N1-14). Levels asserted on every build; 1k/2 nF (1 us) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT).
NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 8
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /07 Cell port C1/
File: 07_cell1.kicad_sch
Title: CTRL-C2000 control card - Cell port C1

Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 8/15

[illegible]

NTC_N returns to AGND here; PCT/EN pairs go to the chip sheet.

TPS26600 Mode operation circuit breaker (CB) [130 – 161 A (RILIM) 8.06k 1 % eq. 4 + table 7.5], 4 ms, 540 ms auto-retry, factory UVLO 15 V / OVP 32.6 V (UVLO, OVP to RTIN): SHDN open = on (self-biased 2.7 V), dV/dt 68k (CSR-03): 0.15–0.23 V/ms, 26.4 V in 115–208 ms: 600 uF draws < 0.137 A, so start-up <= 1.14 A per device and 4.5 A for the whole board (1000 uF <= 4.69 A). Start-up dissipation <= 6.7 W (resistive-load mode, eq. 22)–24 W or 51V50k2k figure 55 at 63 C (graph reading, not verified). Assumed on every ball A shorted cable trim only, circuit life: A PTC was selected: NF-NSMF110 holds only 0.75 A at 60 C.

[illegible]

PWMk (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs: ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = C2_DRV_EN, G_high: outputs high impedance unless CHB_OK, C2_PWM_MCU and RDI are high (INT-15). ePWM5-8 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

OPA4388BIPWR

Difference amp per pair: $V_{\text{adc}} = V_{\text{MD}} 1.25 \text{ V} + G \times (A_{\text{NL}_P} - A_{\text{NL}_N})$, $R_{\text{in}} 10.0\text{k } 0.1\%$

[illegible][illegible]

2xVDD

CB11 1n 50V

AGND

R817 10.0k

C2_AN2_P

C2_AN2_N1

R818 4.02k

VMID

R819 10.0k

C2_AN2_INV

C2_AN2_N

R820 4.02k

C2_AN2_DA

R821 4.7k

C2_AN2_INV

C2_AN2_DA

R822 82p 50V

C2_AN2_DA

DB02

BAT

AD9229

+3V3A

[illegible]

Figure 1: Schematic diagram of the power and ground connections for the LVC1A17. The diagram shows four pins: C2_RDY_P, R827, C2_ID, and C2_NTC_P. C2_RDY_P is connected to GND via a 100k resistor (R827). C2_ID is connected to GND via a 1nF capacitor (CB14). C2_ID is also connected to AGND via a 100nF capacitor (CB15). C2_NTC_P is connected to AGND via a 100nF capacitor (CB16). A separate section shows the connection of C2_RDY_F to R828 (1k) and then to C2_RDY_F. Below this, a section shows the connection of VBIAS to R829 (10.0k) and then to C2_ID, and VBIAS to R830 (10.0k) and then to C2_NTC_P. The text below the diagram explains the push-pull from the power board (PVCELL-25 rev B, DAB-D60 rev B; LVC1A17; Ioff = high impedance w/ unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (NNT-14). Levels asserted on every valid 3V(1) NF (1 us) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS; NTC reads -2.5 V at 25 C.

RDY: push-pull from the power board (PVCCELL-25 rev B, DAB-D60 rev B; LVC1G17, Ioff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build; 1k/1 nF (1 μ s) filter into the GPIO and the driver-enable AND. ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pin): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Pinout diagram of the 40-pin J901 connector for the 6120A021621 module. The diagram shows two rows of pins, 1-20 on the left and 21-40 on the right. Pin 1 is C3_24V, Pin 2 is C3_24V, Pin 3 is GND, Pin 4 is GND, Pin 5 is C3_PWM1_P, Pin 6 is C3_PWM1_N, Pin 7 is C3_PWM2_P, Pin 8 is C3_PWM2_N, Pin 9 is C3_PWM3_P, Pin 10 is C3_PWM3_N, Pin 11 is C3_PWM4_P, Pin 12 is C3_PWM4_N, Pin 13 is GND, Pin 14 is GND, Pin 15 is C3_PWM5_P, Pin 16 is C3_PWM5_N, Pin 17 is C3_PWM6_P, Pin 18 is C3_PWM6_N, Pin 19 is C3_PWM7_P, Pin 20 is C3_PWM7_N, Pin 21 is C3_PWM8_P, Pin 22 is C3_PWM8_N, Pin 23 is GND, Pin 24 is GND, Pin 25 is C3_FET_P, Pin 26 is C3_FET_N, Pin 27 is C3_EN_P, Pin 28 is C3_EN_N, Pin 29 is C3_RDY, Pin 30 is C3_ID, Pin 31 is C3_AN1_P, Pin 32 is C3_AN1_N, Pin 33 is C3_AN2_P, Pin 34 is C3_AN2_N, Pin 35 is C3_AN3_P, Pin 36 is C3_AN3_N, Pin 37 is AGND, Pin 38 is AGND, Pin 39 is C3_NTC_P, Pin 40 is GND. The diagram also shows two capacitors, C901 (10u 50V) and C902 (100n 100V), connected to the C3_24V pins.

6120A021621

NTC_N returns to AGND here; PCT/EN pairs go to the chip sheet.

[illegible][illegible]

U904

OPA4388IPWR

Pin 1: OUTA

Pin 2: C3_AN1_INV

Pin 3: C3_AN1_INI

Pin 4: C3_AN2_INV

Pin 5: C3_AN2_INI

Pin 6: C3_AN3_INV

Pin 7: OUTB

Pin 8: OUTC

Pin 9: C3_AN3_INV

Pin 10: C3_AN3_INI

Pin 11: V-

Pin 12: C3_SPARE_D

Pin 13: C3_SPARE_D

Pin 14: OUTD

Pin 15: C3_AN2_OA

Pin 16: C3_AN1_OA

Power Supply:

- +3V3A
- 100nF 50V (C907)
- AGND

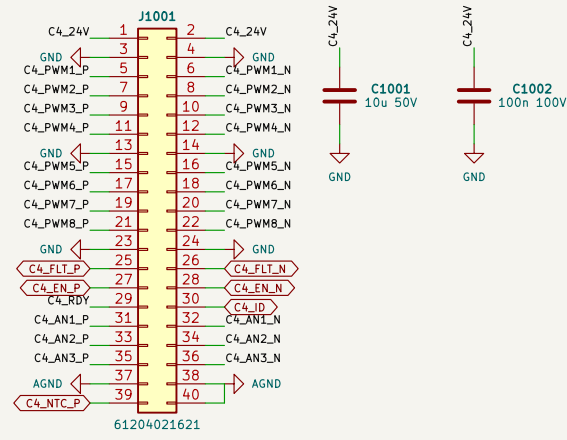
Difference map per unit: Vadc = 125 mV ± 9.6 nV (C/Gc; ADP-charge), RN 10.0 kΩ ± 10 ppm/kΩ, R = 0.02%/100 kΩ, 0.402 V (RT) RR; anti-alias pole 483 Hz, hrf 1 nF C/GC; ADP-charge backplate (ACQPS = 30 ms at 12 bit).
BATS: 125 mV ± 9.6 nV (C/Gc; AGND), all 100 pF, 100 nA, 100 μs, 100 ns, 100 ps, 100 fs, 100 as.
Accepts on any pin differential; on a 1.24–1.49 V common mode. Legs 0.1–2.6 V (ADC span ~3.1 V),
or 0–3.0 V single-ended with ALN = AGND at the same temperatures → 1.25–2.46 V, Op-amp inputs 0.9–1.75 V
(BATS) 0.9–1.75 V (ALN) 0.9–1.75 V (AGND) keep it up! Keep it up!
Linear + indicator current (NI=12, one constant); FCVELL=25 LEM LA=150 P=200 MVA 1.25 V C=1 (source,
linear + NI=14 → 8.04 mA at the CGC, 0.0759 A/LIN; CMPPS: backgate ~91.5 A, The LA 150 – P offset (up to
~2.4 V), 150 V, 150 A, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W, 150 W,
DAB-D00 (C/C2C); ANI 5.005 mA/V (+/-1.58 mV, Vdc ≈ 2.79 V, ANI ≈ 2.75 V single-ended, 49.9 A
currents (+0.5 % C/gc). Any ANI carries the open-input bias. Channel D: VMIDT output.

[illegible][illegible][illegible]

RDY: push-pull from the power board (PVCILL-25 rev B, DAB-D68 rev B; LVC1617, loff = high impedance when unpowered) or open drain with 1k ohm pull-up: the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (NT-14). Levels asserted on every build: 14/1 (1 u) filter into the GND and the 5V available AND: 10.0k from VBIAS (5V) decade table in the pin plan (MUX0DUT).

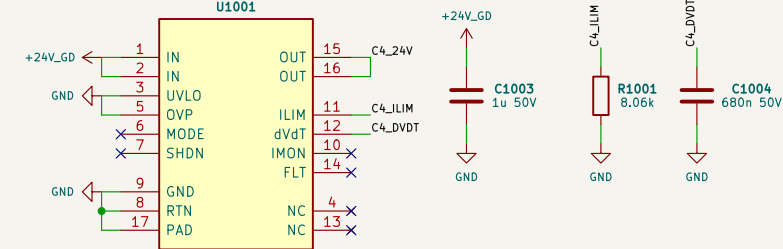
NTC (floating part): 10.0k from VBIAS, 10K NTC reads 1.25 V at 25 C.

C4 CONNECTOR (CELL CONTRACT)



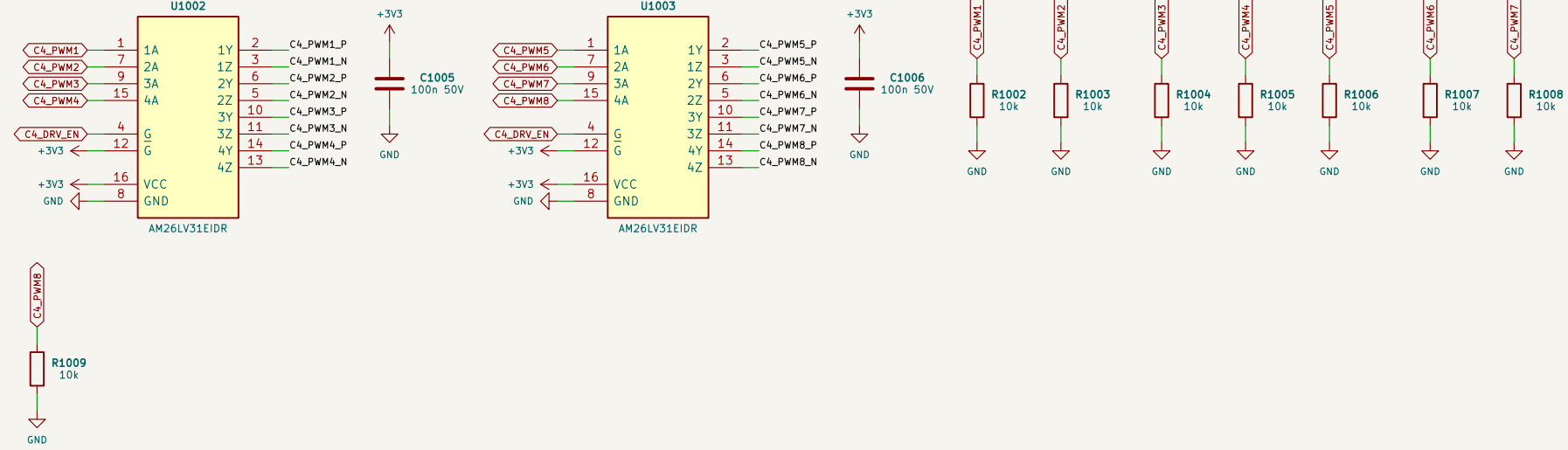
Pin map = Interfaces.CELL. Pins 1-2 carry +24V_GD only (gate-driver bias, channel B switched).
NTC_N returns to AGND here. FLV/EN gains go to the trip sheet.

+24V_GD EFUSE FOR C4 (<= 1.0 A CONTINUOUS, <= 600 uF)



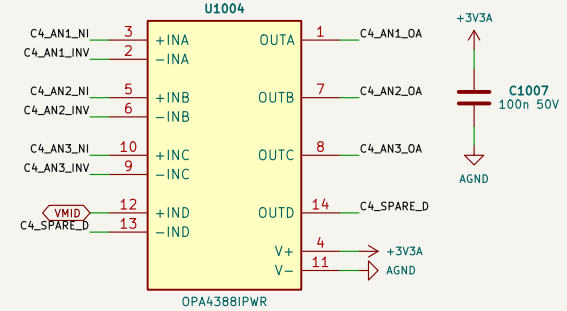
TPS26600, MODE open: circuit breaker I(CB) 1.30-1.61 A (RILIM 8.06k 1 %, eq. 4 + table 7.5), 4 ms, 540 ms auto-retry; factory UNLO 15 V / OVP 32.6 V (UVLO, OVP to RTN); SHDN open = on (self-biased 2.7 V).
dVdI 680k (CSR-D3): 0.13-0.23 V/ms, 26.4 V in 115-208 ms; 600 uF draws <= 0.137 A, so start-up <= 1.14 A per port and <55 A for four (SYS <24V_GD limit <= 4.69 A). Start-up dissipation <= 6.7 W (resistive-load model, eq. 22-24) vs SLVSDG2G figure 55 at 85 C (graph reading, not verified). Asserted on every build.
A shorted cell cable trips only its own eFuse. A PTC was rejected: MF-MSMF110 holds only 0.73 A at 60 C.

PWM LINE DRIVERS (EPWM13-16)



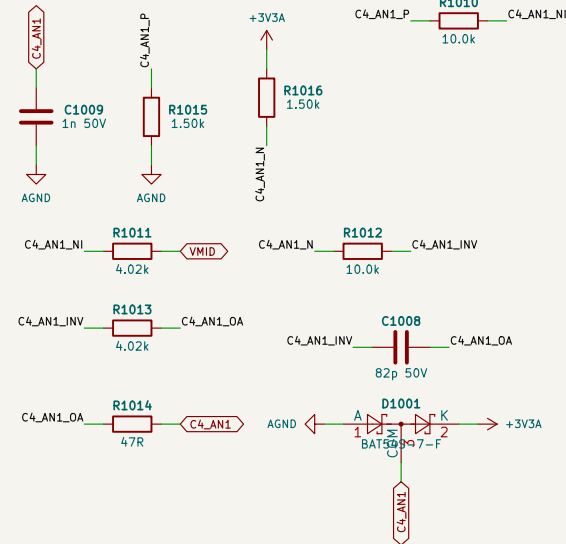
PWMs (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-8), so every gate command is 0 until firmware runs.
G = C4_DRV_EN, G- high; outputs high impedance unless CHB_OK, C4_EN_MCU and RDY are high (INT-15).
ePWM13-16 have no high-resolution PWM; buck-boost cells only, not a DAB bridge.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

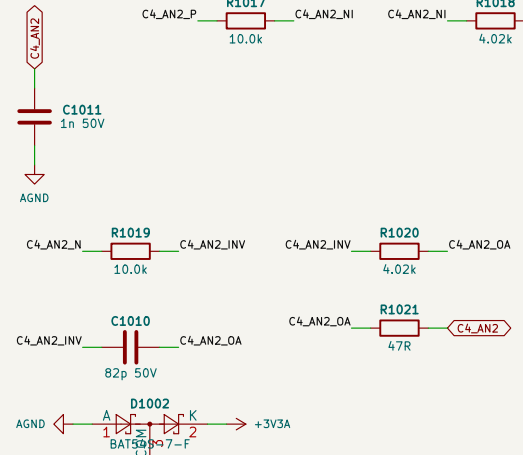


Difference amp per pair: $V_{\text{dsc}} = V_{\text{MID}} 1.25 \text{ V} + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$.
 $C_f || R_f$ anti-alias pole 483 kHz, 47 R + 1 nF COG; ADC charge bucket (ACQPS >= 320 ns at 12 bit).
BAT5AS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated.
Accepts on any AN pair differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with $ANL_N = AGND$ at the source (temperatures) -> 1.25-2.46 V. Op-amp inputs 0.9-1.75 V (RRIO). Source resistance adds to the 10.0k; keep $R_s <= 100 \text{ R}$ (1 % gain) or calibrate it out.
AN1 = inductor current (INT-12, one constant); PVCELL-25 LEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11.4 A -> 8.0k mV/A at the ADC. 0.0759 A/LSD; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it.
DAB-D60 (C1/C2): AN1 5.005 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMID follower.

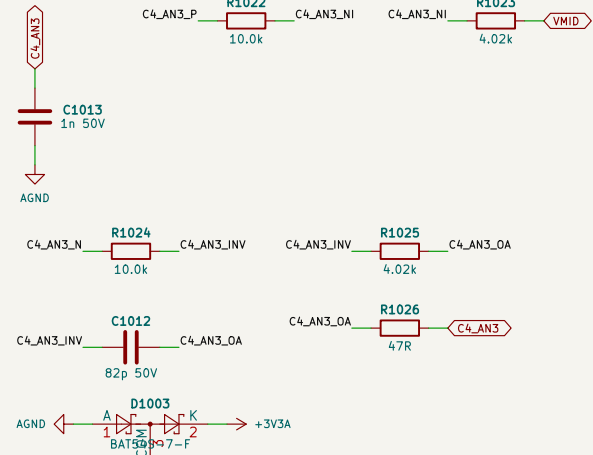
C4_AN1 -> ADCIND0 / CMPIN7P



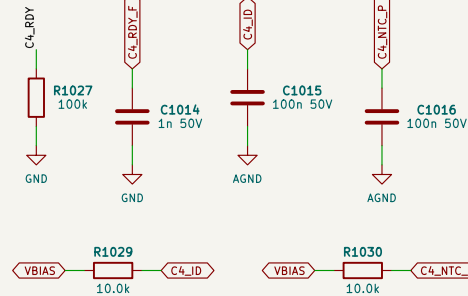
C4_AN2 -> ADCINA5



C4_AN3 -> ADCINB4



RDY, ID, NTC



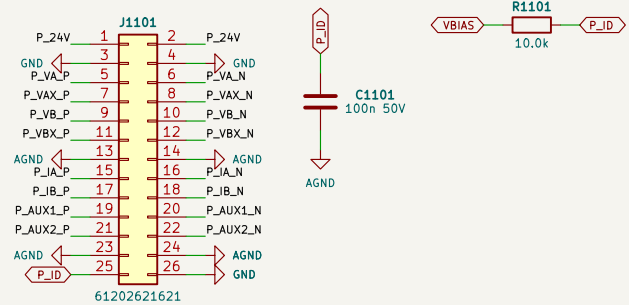
RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build: 1k/1 nF (1 us) filter into the GPIO and the driver-enable AND, ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT).
NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 11
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /10 Cell port C4/
File: 10_cell4.kicad_sch
Title: CTRL-C2000 control card - Cell port C4

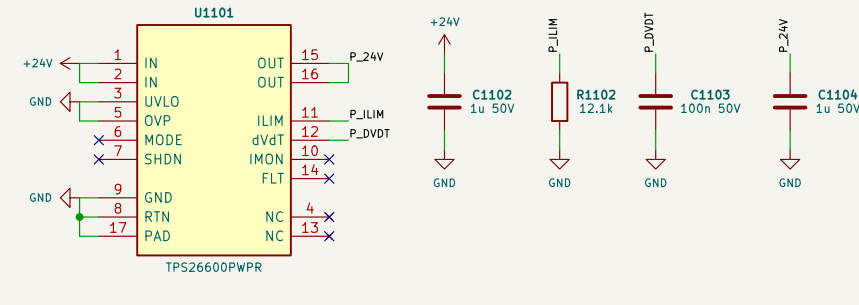
Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 11/15

PORT-BOARD CONNECTOR (PORT CONTRACT)



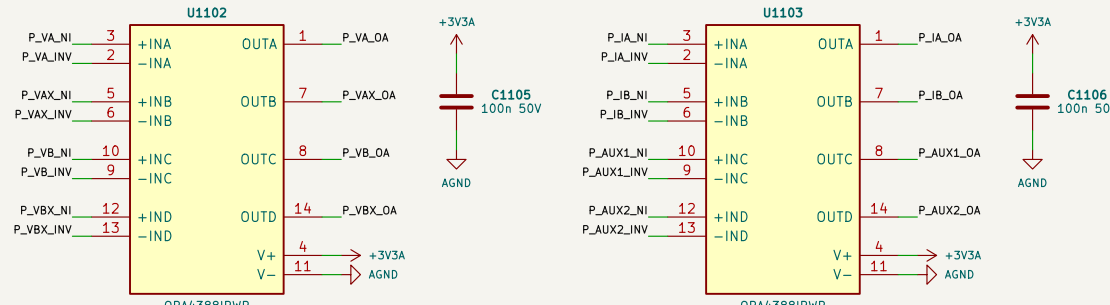
Pin map = Interfaces:PORT. P_ID: 10.0k from VBIAS. The port board sums its hold readbacks onto ID: decode table (both codes x 9 states, unpowered, open, short; every band >= 8 counts apart) in the pin plan. An unpowered board reads its own band inside the PV-PORT PPB window: firmware decodes it, and in hardware VA/VB read <= -161 V and trip (INT-07, asserted).

P_24V EFUSE (INT-06, CSR-12)



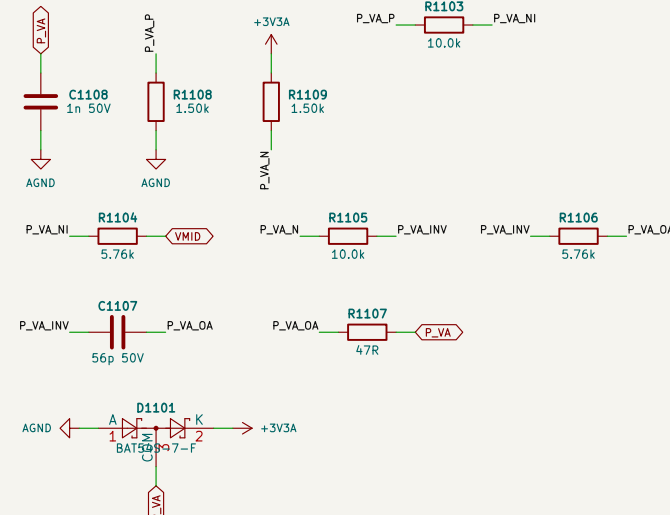
Port board: up to 0.7 A for up to 200 s (battery port: two hold-closed coils + sensors), TPS26600. MODE open: circuit breaker I(CB) 0.851-1.075 A (RILIM 12.1k 1 %), so the hold path never trips it; a shorted port board or cable trips only this branch (4 ms, 540 ms retry), and CTRL 0.35 A + 1.08 A stays below the SYS +24V efuse (>= 1.82 A); the control card keeps running. dV/dt 100n: 0.86-1.56 V/ms into 5.3 uF. Factory UVLO 15 V / OVP 32.6 V; SHDN open. The PTC (0.73 A hold at 60 C) is gone.

ANALOG FRONT ENDS (2 X OPA4388)

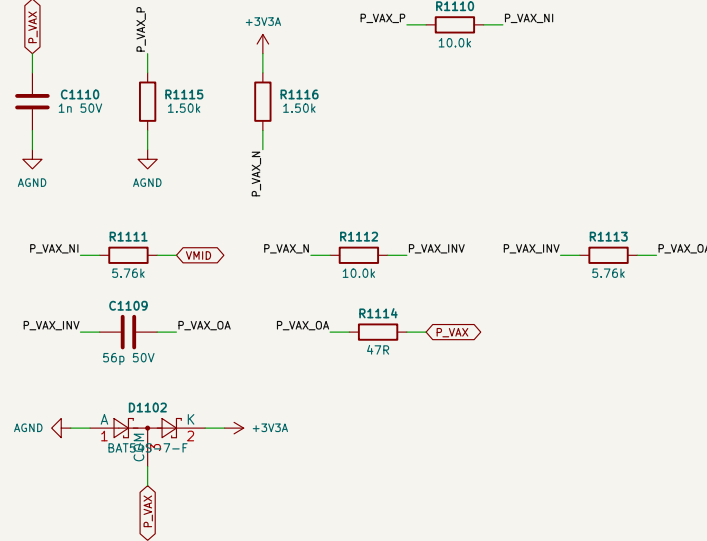


Difference amp per pair: $V_{adc} = V_{MID} 1.25 V + G \times (AN_P - AN_N)$, $R_{in} 10.0k 0.1 \% 10 ppm/K$, $G = 5.76k/10.0k = 0.576$; $+/-2 V \rightarrow 0.10-2.40 V$. Cf [1] Rf anti-alias pole 493 kHz, 47 R + 1 nF CG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT54S clamps the ADC node to 3V3A/AGND. All values calculated, not simulated. Open-input bias (INT-07): 1.50k $AN_P \rightarrow AGND$ and $AN_N \rightarrow +3V3A$, so an open conductor or ribbon reads far outside the trip window (instead of 0 (worst readings: check trips build line)). Loads the isolated amplifier with -1.3 k (AMC3330/AMC3302 RLOAD >= 1 k); source + cable resistance R gives a gain error $R/1.5 k (+0.07 \% \text{ for } 1 R)$, removed by the gain calibration. VA, VB, IB, IB sit on CMPS5 positive inputs: trip paths and bands in the pin plan.

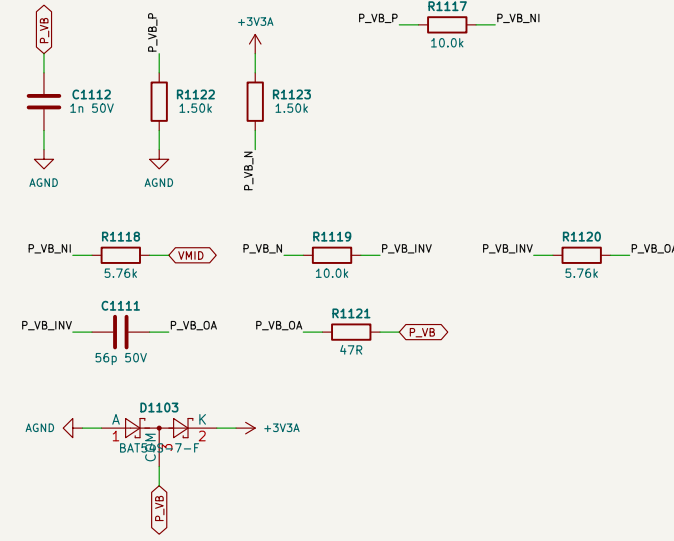
P_VA --> ADCIND2 / CMPIN8P



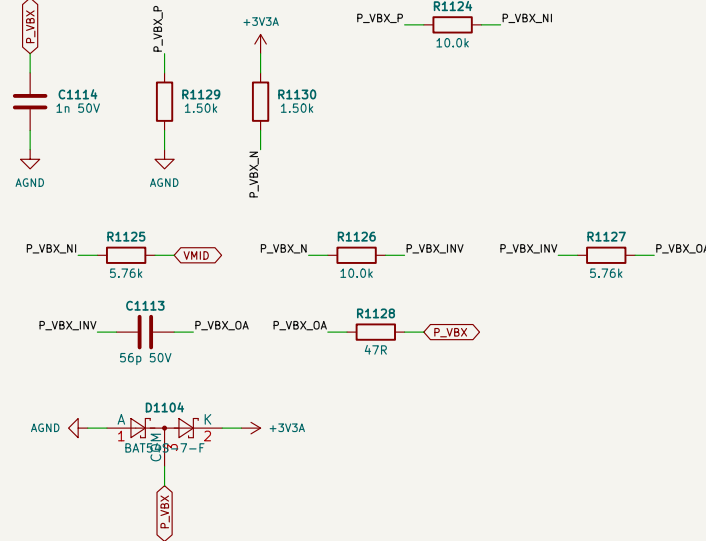
P_VAX --> ADCINA0



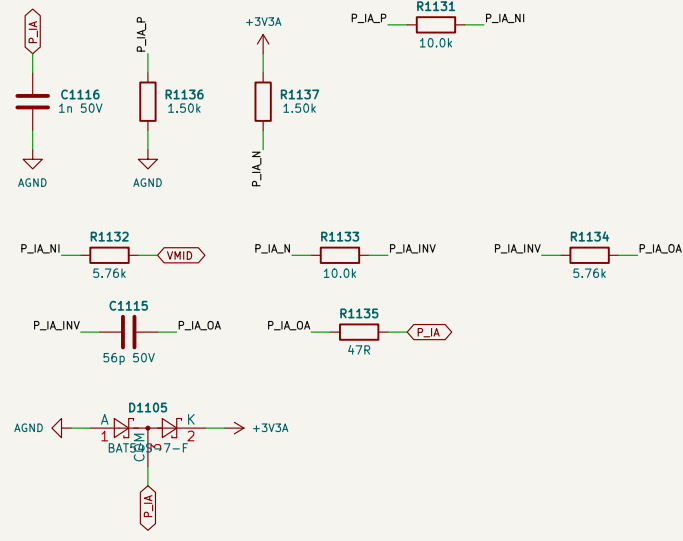
P_VB --> ADCIN14 / CMPIN4P



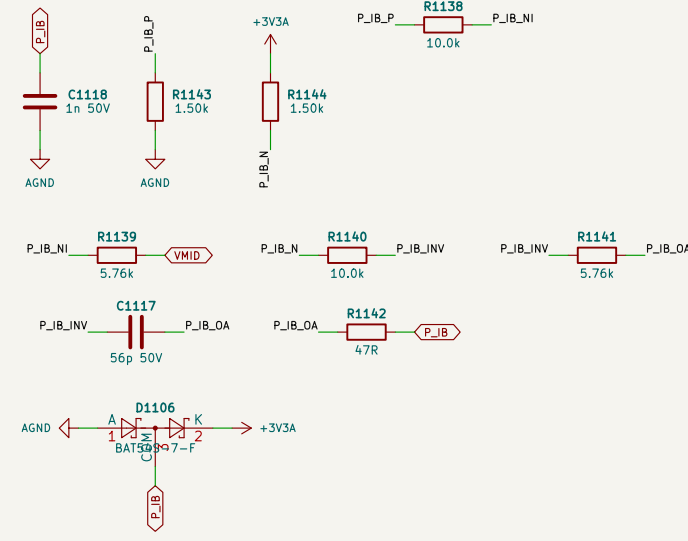
P_VBK --> ADCIN15



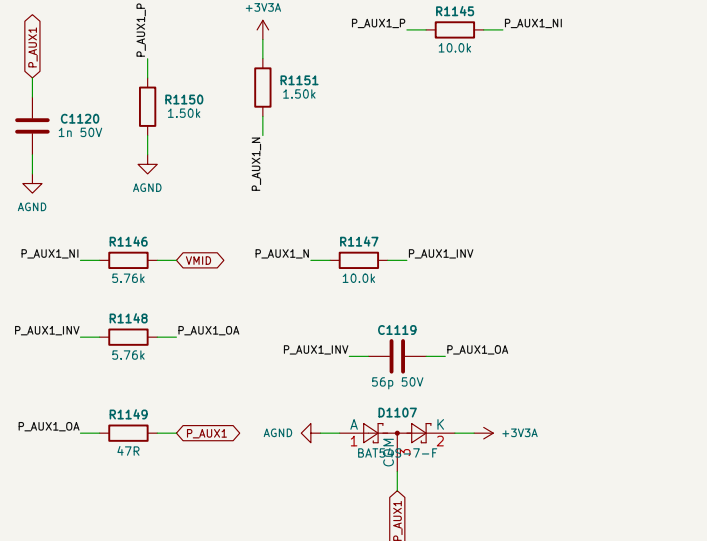
P_JA --> ADCINA4 / CMPIN2P



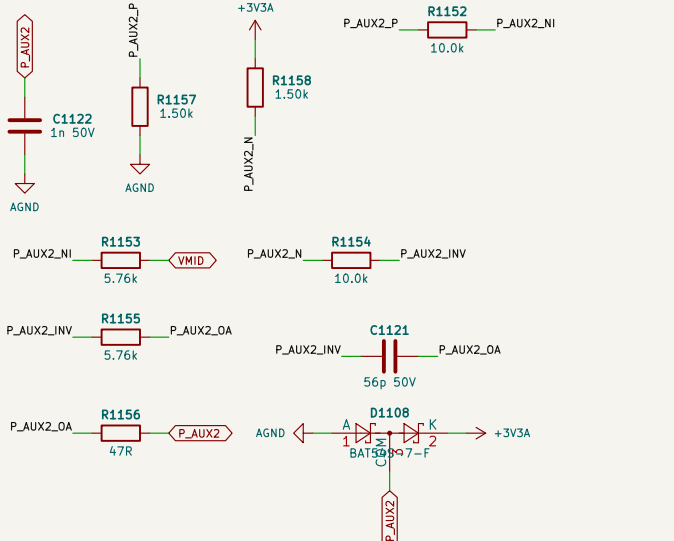
P_JB --> ADCINC4 / CMPIN5P



P_AUX1 --> ADCINA1



P_AUX2 --> ADCINB1



Not bench-validated. Values marked calculated are not simulated.
Sheet 12
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

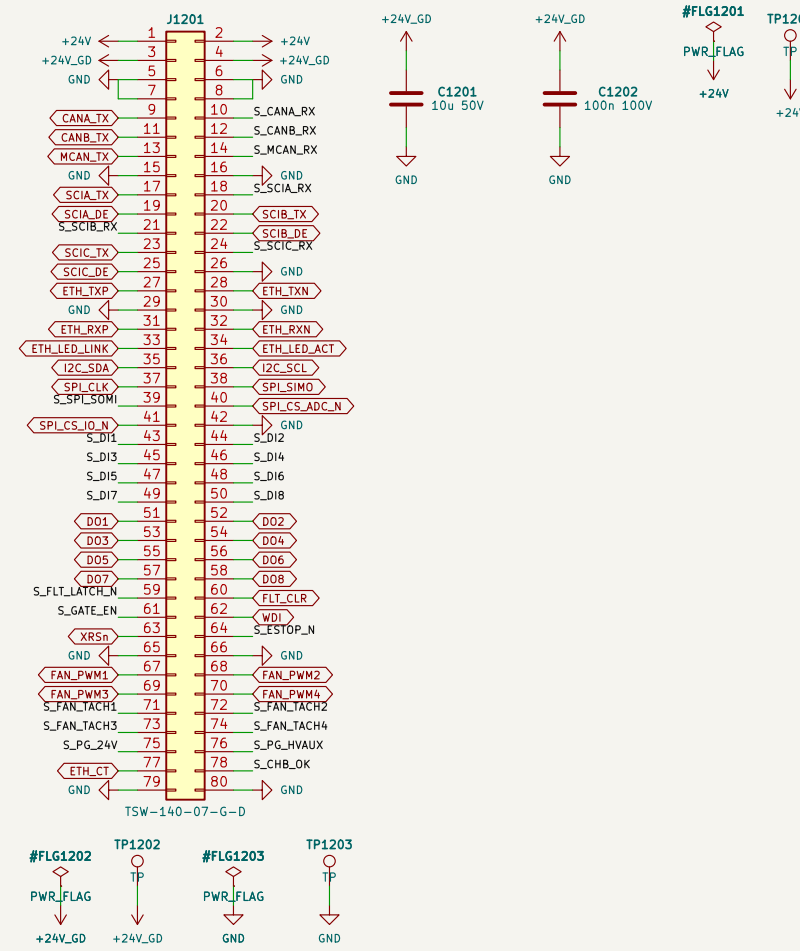
Sheet: /11 Port-board analog inputs/
File: 11_port.kicad_sch

Title: CTRL-C2000 control card - Port-board analog inputs

Size: A2 Date: 2026-10-04 Rev: E0

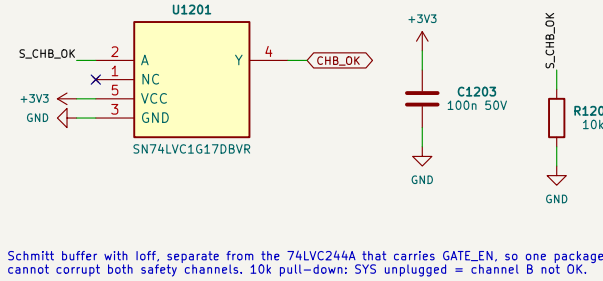
KiCad E.D.A. 10.0.6 Id: 12/15

SYS CONNECTOR (SYS CONTRACT)



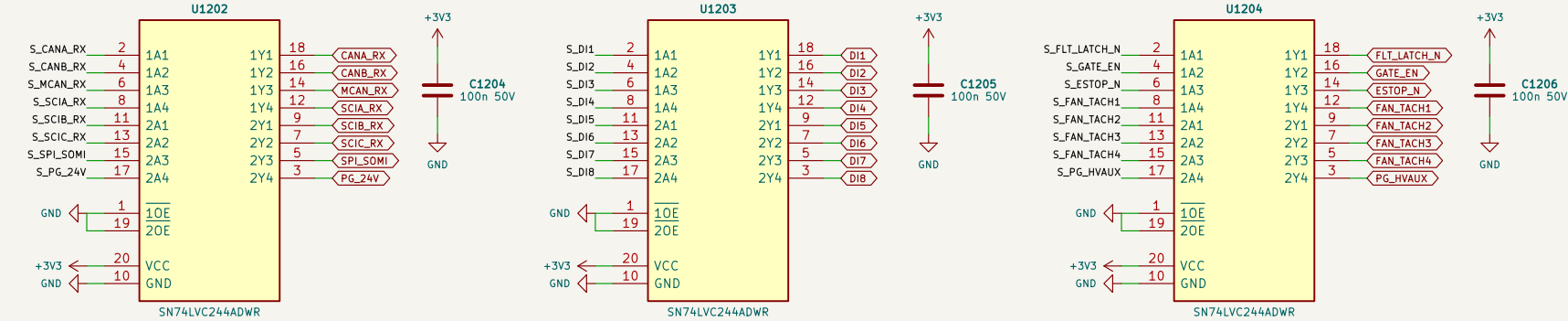
Pin map = interfaces.SYS: SYS_RST_N lands directly on XRSn (open drain both sides).
 +24V (pins 1-2): this card's regulators and the PORT connector. +24V_GD (pins 3-4): switched by safety channel. 18 on SYS: goes ONLY to the four cell eFuses (cell sheets). The two never meet here. ETH_CT (77): PHY analog supply to the magnetics centre taps (Ethernet sheet).
 Inputs from SYS arrive as `S_<name>` and reach the MCU as `<name>` through the buffers.

CHB_OK BUFFER (CHANNEL B, OWN PACKAGE)



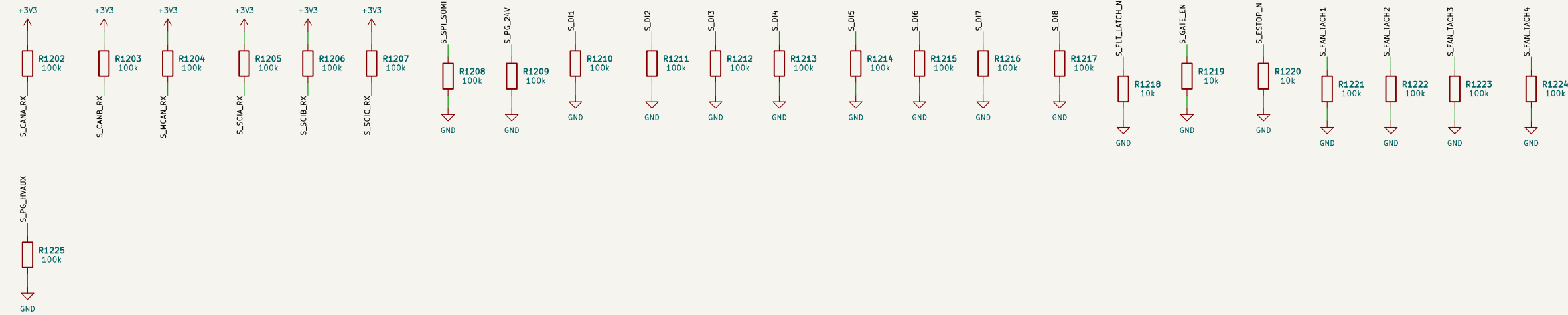
Schmitt buffer with 10k, separate from the 74LVC244A that carries GATE_EN, so one package cannot corrupt both safety channels. 10k pull-down: SYS unplugged = channel B not OK.

SYS INPUT BUFFERS (IOFF)



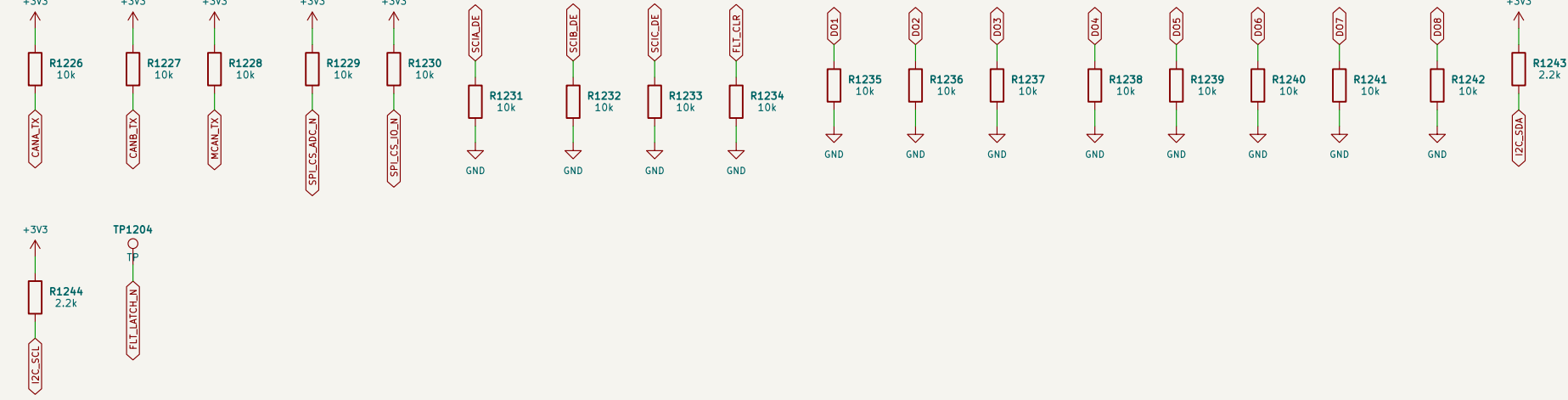
SYS makes its own 3.3 V and may drive these lines before this card's VDDIO is up; SPRSP14E 7.1 allows ± 20 mA total input-clamp current and 7.10.1.4.2 forbids driving pins early. 74LVC244A inputs and outputs are high impedance unpowered (loff) and its outputs never exceed VDDIO. tpd -5 ns.

INPUTS WITH SYS UNPLUGGED



FLT_LATCH_N, GATE_EN, ESTOP_N: 10k to GND, so an unplugged SYS reads as tripped and the gates stay off.
Other inputs: 100k to their idle level (CAN/SCI receive high, the rest low).

OUTPUTS WHILE THE MCU IS IN RESET



GPIOs float in reset: CAN TX recessive, RS-485 DE off, DO and FLT_CLR low, SPI chip selects high
I2C: 2.2k pull-ups here (bus master side); SYS should not add a second pair.

Not bench-validated. Values marked calculated are not simulated.
Sheet 13
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /12 SYS-I0-AUX connector/
File: 12_sys.kicad_sch

Title: CTRL-C2000 control card - SYS-IO-AUX connector

Size: A2

Date: 2026-10-04

Rev: E0

1.6

Id: 13/15

JTAG CUI-20 (SPRU655I TABLE 13)

Diagram illustrating the JTAG pin connections for the FTSH-110-01-L-DV-K device. The device has 20 pins, numbered 1 through 20. The connections are as follows:

- Pin 1:** TMS (2.2k pull-up to +3V3)
- Pin 2:** TRStn (2.2k pull-down to GND)
- Pin 3:** JTAG_VREF (100R pull-up to +3V3)
- Pin 4:** GND
- Pin 5:** TDO (4.7k pull-up to +3V3)
- Pin 6:** GND
- Pin 7:** TCK (4.7k pull-up to +3V3)
- Pin 8:** GND
- Pin 9:** JTAG_EMU0 (4.7k pull-up to +3V3)
- Pin 10:** GND
- Pin 11:** JTAG_EMU1 (4.7k pull-up to +3V3)
- Pin 12:** GND
- Pin 13:** XRSn (4.7k pull-up to +3V3)
- Pin 14:** GND
- Pin 15:** GND
- Pin 16:** GND
- Pin 17:** GND
- Pin 18:** GND
- Pin 19:** GND
- Pin 20:** GND

The device is labeled FTSH-110-01-L-DV-K.

TMS 2.2k pull-up and TRStn 2.2k pull-down (SPRSP14E table 6-1); TDIS to GND; VTRef via 100 R; RTCK looped to TCK; EMU0/1 4.7k pull-ups (no EMU pins on F2836x); RESET (open drain) on XRSn; EMU2-EMU4 (17-19) grounded as SPRSP14E 7.10.7 requires (CSR-07).

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Pin 1 GND, 2 TX (from MCU), 3 RX (to MCU); 100 R series each way.

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CALIBRATION / SERIAL-NUMBER EEPROM

U1401
24LC256T-1/SN

+3V3

8 VCC
1 A0
2 A1
3 A2
7 WP
4 VSS

EEPROM_WP
GND

5 SDA
6 SCL

I2C_SDA
I2C_SCL

+3V3

C1401
100n 50V

GND

+3V3

R1408
10k

EEPROM_WP

Address 1010111 (A2-A0 high = 0x57) so it cannot collide with a SYS EEPROM at the default 0x50.
WP pulled high: write-protected unless firmware drives EEPROM_WP low.

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WP pulled high: write-protected unless firmware drives EEPROM_WP low.

STATUS LEDs AND ERRORSTS

The diagram illustrates the circuit for STATUS LEDs and ERRORSTS. It features five LEDs (D1401-D1405) connected to +3V3 through resistors (R1409-R1413) and to GND. The LEDs are labeled LED_PWR_A, LED_CPU1_A, LED_CPU2_A, LED_CM_A, and LED_FLT_A. The ERRORSTS pin is connected to +3V3 through resistor R1414 and to GND through a pull-down resistor. A TP1401 test point is also shown.

GPIO sinks about 2 mA (680 R). ERRORSTS needs an external pull-down.

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SDFM EXPANSION HEADER (NO MATING CONNECT YET)

The diagram shows the SDFM Expansion Header (J1403) with 20 pins. The connections are as follows:

- Pin 1:** +3V3
- Pin 2:** GND
- Pin 3:** SD_CLK_OUT
- Pin 4:** GND
- Pin 5:** SD_CLK_RET
- Pin 6:** GND
- Pin 7:** SD1_D1
- Pin 8:** SD1_D2
- Pin 9:** SD1_D3
- Pin 10:** SD1_D4
- Pin 11:** GND
- Pin 12:** GND
- Pin 13:** SD2_D1
- Pin 14:** SD2_D2
- Pin 15:** SD2_D3
- Pin 16:** SD2_D4
- Pin 17:** GND
- Pin 18:** GND
- Pin 19:** +3V3
- Pin 20:** GND

The header is labeled J1403 and TSW-110-07-G-D. A separate component R1415 (33R) is shown connected to SD_CLK_OUT and SD_CLK_RET.

8 modulator data inputs (SD1/SD2 D1-D4) and one clock: XCLKOUT out, returned clock to all eight SDx_Cy inputs. 3.3 V CMOS; for an adjacent sensing board only.

8 modulator data inputs (SD1/SD2 D1–D4) and one clock: XCLKOUT out, returned clock to all eight SDx_Cy inputs. 3.3 V CMOS; for an adjacent sensing board only.